Jeonghyeok Do

Video and Image Computing Lab, KAIST

Google Scholar | jeonghyeokdo.github.io
ehwjdgur0913@kaist.ac.kr | jeonghyeok-do



RESEARCH EXPERIENCE

03.2026 – Present
Principal Investigator, Sejong Science Fellowship
Project: AI-based Satellite Image Visualization and Analysis
Funded by National Research Foundation of Korea (NRF) and MSIT

09.2025 – Present
Postdoctoral Researcher, KAIST
Information & Electronics Research Institute

03.2019 – 08.2025
Research Assistant, KAIST
Various government- and industry-funded projects
Agency for Defense Development, LG Electronics, SK Hynix, Stellarvision

EDUCATION

03.2021 – 08.2025
Ph.D. in Electrical Engineering, KAIST
Advisor: Prof. Munchurl Kim (VIClab)
Thesis: *A Study on Efficient Skeletal-Temporal Transformer for Skeleton-based Action Recognition*
GPA 4.21/4.30 — 40 credits completed (include M.S.)

03.2019 – 02.2021
M.S. in Electrical Engineering, KAIST
Advisor: Prof. Munchurl Kim (VIClab)
Thesis: *Effective Labeling on Unlabeled Data for Imbalanced Training Dataset*
GPA 4.20/4.30 — 51 credits completed

03.2015 – 02.2019
B.S. in Electrical Engineering, KAIST
Double Major: Mathematical Sciences
GPA 3.80/4.30 — 151 credits completed, *Cum Laude*

RESEARCH PUBLICATIONS

International Conferences

1. **J. Do** and M. Kim, "GeoCore-9B: Towards Geo-Aware Generative Foundation Models in Earth Observation," in *Advances in Neural Information Processing Systems (NeurIPS)*, Dec. 2026.
2. **J. Do**, Y. Chen, G. Youk, and M. Kim, "Less is More: Compact-Token Masked Feature Prediction for Skeleton Representation Learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, Dec. 2026.
3. G. Youk, **J. Do**, D. Kim, J. Oh†, and M. Kim†, "AVSR-Diff: Scale-Agnostic Diffusion Priors for Temporally Consistent Arbitrary-Scale Video Super-Resolution," in *European Conference on Computer Vision (ECCV)*, †Co-corresponding authors, Sep. 2026.
4. **J. Do** and M. Kim, "Bridging the Skeleton-Text Modality Gap: Diffusion-Powered Modality Alignment for Zero-shot Skeleton-based Action Recognition," in *International Conference on Computer Vision (ICCV)*, Oct. 2025.
5. **J. Do**, S. Kim, G. Youk, J. Lee†, and M. Kim†, "PAN-Crafter: Learning Modality-Consistent Alignment for PAN-Sharpening," in *International Conference on Computer Vision (ICCV)*, †Co-corresponding authors, Oct. 2025.
6. S. Kim, **J. Do**, J. Lee†, and M. Kim†, "U-Know-DiffPAN: An Uncertainty-aware Knowledge Distillation Diffusion Framework with Details Enhancement for PAN-Sharpening," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, †Co-corresponding authors, Jun. 2025.
7. **J. Do** and M. Kim, "SkateFormer: Skeletal-Temporal Transformer for Human Action Recognition," in *European Conference on Computer Vision (ECCV)*, Oct. 2024.
8. **J. Do** and M. Kim, "MorphVAD: Efficient Video Anomaly Detection Using Morphological Transformation," in *International Conference on Visual Communications and Image Processing (VCIP)*, IEEE, Dec. 2023.
9. **J. Do** and M. Kim, "Multi-modal Transformer for Indoor Human Action Recognition," in *2022 22nd International Conference on Control, Automation and Systems (ICCAS)*, IEEE, Nov. 2022, pp. 1155–1160.
10. C. O. Ancuti, C. Ancuti, F.-A. Vasluianu, and R. Timofte [and 48 others, including **J. Do**], "NTIRE 2020 Challenge on Non-Homogeneous Dehazing," in *Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, Jun. 2020.

International Journals

11. **J. Do**, J. Lee, S. Lee, and M. Kim, "C-DiffSET: Leveraging Latent Diffusion for SAR-to-EO Image Translation with Confidence-Guided Reliable Object Generation," *IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)*, Jun. 2026.

12. **J. Do**, S. Lee, and M. Kim, "ReFlowSET: Representation-Aligned Latent Flow Matching for SAR-to-EO Image Translation," *IEEE Geoscience and Remote Sensing Letters (GRSL)*, Nov. 2026.
13. S. Kim, **J. Do**, and M. Kim, "Pseudo-Supervised Learning for Semantic Multi-Style Transfer," *IEEE Access*, vol. 9, pp. 7930–7942, Jan. 2021.
14. S. Ki, **J. Do**, and M. Kim, "Learning-based JND-directed HDR Video Preprocessing for Perceptually Lossless Compression with HEVC," *IEEE Access*, vol. 8, pp. 228 605–228 618, Dec. 2020.

Preprints

15. **J. Do** and M. Kim, "GeoCR: Learning a Generalist Cloud Removal Prior from Heterogeneous Observations," in *arXiv preprint*, Sep. 2026.
16. **J. Do** and M. Kim, "GeoSET: Generalist Foundation Model for SAR-to-EO Image Translation," in *arXiv preprint*, Sep. 2026.
17. **J. Do** and M. Kim, "SAREO-FM: Decoupled Semantic Supervision for SAR–EO Foundation Models," in *arXiv preprint*, Oct. 2026.
18. **J. Do**, Y. Chen, and M. Kim, "One for All: Generalist Foundation Model for Cross-Sensor Skeleton Representation Learning," in *arXiv preprint*, Sep. 2026.
19. Y. Chen, M. Kim[†], and **J. Do**[†], "MotionMaestro: Masked Tokenization for Unified Motion Generation," in *arXiv preprint*, [†]Co-corresponding authors, Sep. 2026.

Under Review

20. Y. Chen[*], **J. Do**[*], and M. Kim, "What Makes a Good Motion Tokenizer for 3D Human Motion Generation?" In *Under review*, [*]Equal contribution, 2026.

Domestic Conferences

21. **J. Do**, S. Lee, and M. Kim, "Comparative Analysis of Latent Space Autoencoders for SAR-to-EO Image Translation," in *Korea Society of Remote Sensing (KSRS) Workshops*, Oct. 2026.
22. **J. Do**, S. Lee, and M. Kim, "SAR-to-Optical Domain Translation and Utilization Strategies," in *Korea Institute of Broadcast and Media Engineers (KIBME) Workshops*, Jun. 2026.
23. Y. Chen, **J. Do**, and M. Kim, "Skeleton Aware Augmentation for Skeleton Representation Learning," in *Korea Institute of Broadcast and Media Engineers (KIBME) Workshops*, Jun. 2026.
24. **J. Do** and M. Kim, "Beyond Fixed Partitions: Learning Dynamic Joint Grouping for Skeleton-based Action Recognition," in *Korea Institute of Broadcast and Media Engineers (KIBME) Workshops*, Dec. 2025.
25. **J. Do** and M. Kim, "Priors on Bone Length Consistency for Skeleton-based Action Recognition," in *Korea Institute of Broadcast and Media Engineers (KIBME) Workshops*, Jun. 2023.
26. **J. Do** and M. Kim, "Efficient Video Anomaly Detection with Future Frame Prediction," in *Korea Institute of Military Science and Technology (KIMST) Workshops*, Nov. 2021.
27. **J. Do**, H. Sim, G. Lee, S. Baek, and M. Kim, "Efficient Video Anomaly Detection via Future Frame Prediction and Dual Memory Network," in *Korea Institute of Industrial Engineers (KIIE) Workshops*, Nov. 2021.
28. **J. Do** and M. Kim, "Adaptive Attention based U-Net for Image Dehazing," in *Korea Institute of Military Science and Technology (KIMST) Workshops*, Nov. 2020.
29. **J. Do** and M. Kim, "Wafer Map Defect Pattern Classification with Progressive Pseudo-Labeling Balancer," in *Korea Institute of Broadcast and Media Engineers (KIBME) Workshops*, Nov. 2020.

PEER REVIEW EXPERIENCE

2027	Reviewer — **AAAI**
2026	Reviewer — **BMVC**, **NeurIPS**, **AAAI**, **IJCV**, **TMM**
2025	Reviewer — **ICCV**, **CVPR**, **TIP**, **TNNLS**
2024	Reviewer — **VCIP**
2023	Reviewer — **VCIP**

RESEARCH PROJECT EXPERIENCE

Government-Funded Projects

03.2026 – Present **AI-based Satellite Image Visualization and Analysis** (Principal Investigator)
Sejong Science Fellowship — NRF and MSIT

02.2026 – Present **Remote Sensing Foundation Models & Vision-Language Models** (Project Lead)
Advanced GPU Utilization Support Project — MSIT and NIPA

04.2025 – Present **AI-based Computer Vision Study for Satellite Image Processing and Analysis** (Project Lead)
Ministry of Science and ICT (MSIT)

01.2020 – 12.2022 **MX-24: Soldier/Unit Action Recognition based on Multi-modal Learning** (Project Lead)
Agency for Defense Development (ADD) — CARAI

Industry-Funded Projects

11.2023 – 12.2024 **AI-based SAR-to-EO Image Translation** (Project Lead) Stellarvision

01.2021 – 10.2021 **Video Anomaly Detection based on Image Sequence Prediction** (Project Lead) SK Hynix

01.2020 – 11.2020 **Defect Auto Define with Edge BMW Classification** (Project Lead) SK Hynix

09.2019 – 01.2020 **Low-complexity Deep Learning based Super-Resolution** (Support) LG Electronics

PATENTS

01.2026 **Method and Apparatus for Human Action Recognition**
US Patent Application No. 19/504,068

12.2025 **Method and Apparatus for Human Action Recognition**
PCT Application No. PCT/KR2025/020742

12.2025 **Uncertainty Map-based Image Processing Method and Apparatus**
Korean Patent Application No. 10-2025-0188253

12.2024 **Image Restoration Method and Apparatus using Uncertainty-aware Knowledge Distillation and Frequency Component Enhancement**
Korean Patent Application No. 10-2024-0186497

12.2024 **Skeleton-based Action Recognition Method and Apparatus**
Korean Patent Application No. 10-2024-0186489

12.2024 **Method and Apparatus for Image Translation**
Korean Patent Application No. 10-2024-0186157

TECHNOLOGY TRANSFER

04.2025 **Generative AI-based SAR-to-EO Image Translation Technology**
Transferred to Stellarvision

INVITED TALKS

02.2026 **Deep Learning-based Multi-modal Satellite Image Fusion**
Invited lecture at Prof. Jungho Im's Lab, UNIST

HONORS & AWARDS

03.2021 – 08.2025 **Government Scholarship** for Ph.D. Students

03.2019 – 02.2021 **KAIST Scholarship** for M.S. Students

03.2015 – 02.2019 **National Scholarship** for Undergraduate Students

02.2015 **KAIST President Award**

09.2014 **Hanseong Nobel Scholarship Student**

TEACHING EXPERIENCE

Fall 2023 Teaching Assistant — **EE534** Pattern Recognition

Spring 2023 Teaching Assistant — **EE548** Matrix Computations for Signal Processing

Fall 2022 Teaching Assistant — **EE432** Digital Signal Processing; **EE534** Pattern Recognition

Spring 2022 Teaching Assistant — **EE432** Digital Signal Processing; **EE202** Signals and Systems

Fall 2021 Teaching Assistant — **EE405** Electronics Design Lab.; **EE202** Signals and Systems

Spring 2021 Teaching Assistant — **EE432** Digital Signal Processing

SKILLS

Languages **Korean** (Native), **English** (Conversational)

Coding **Python**, **PyTorch**, **MATLAB**, **LaTeX** (all Proficient)